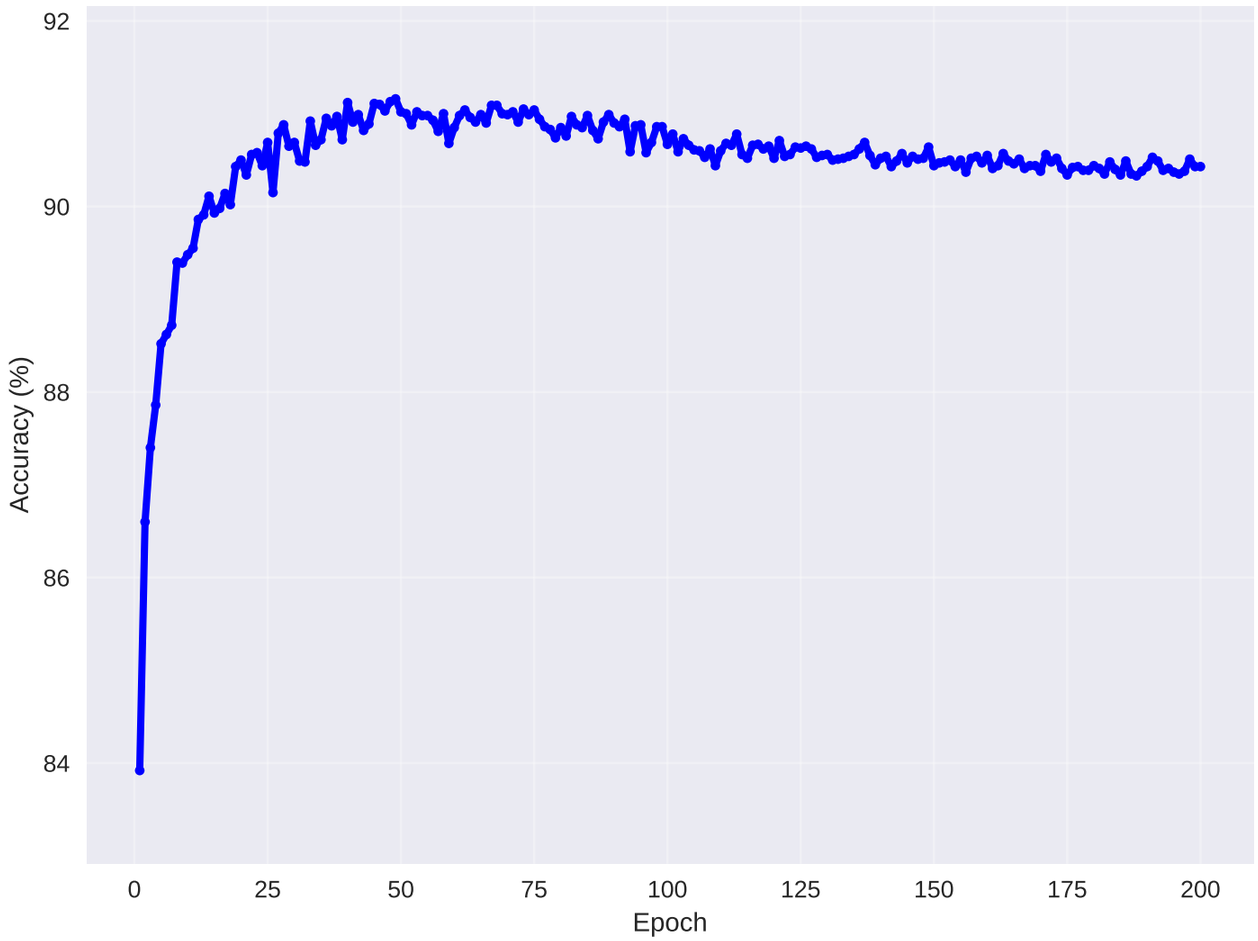
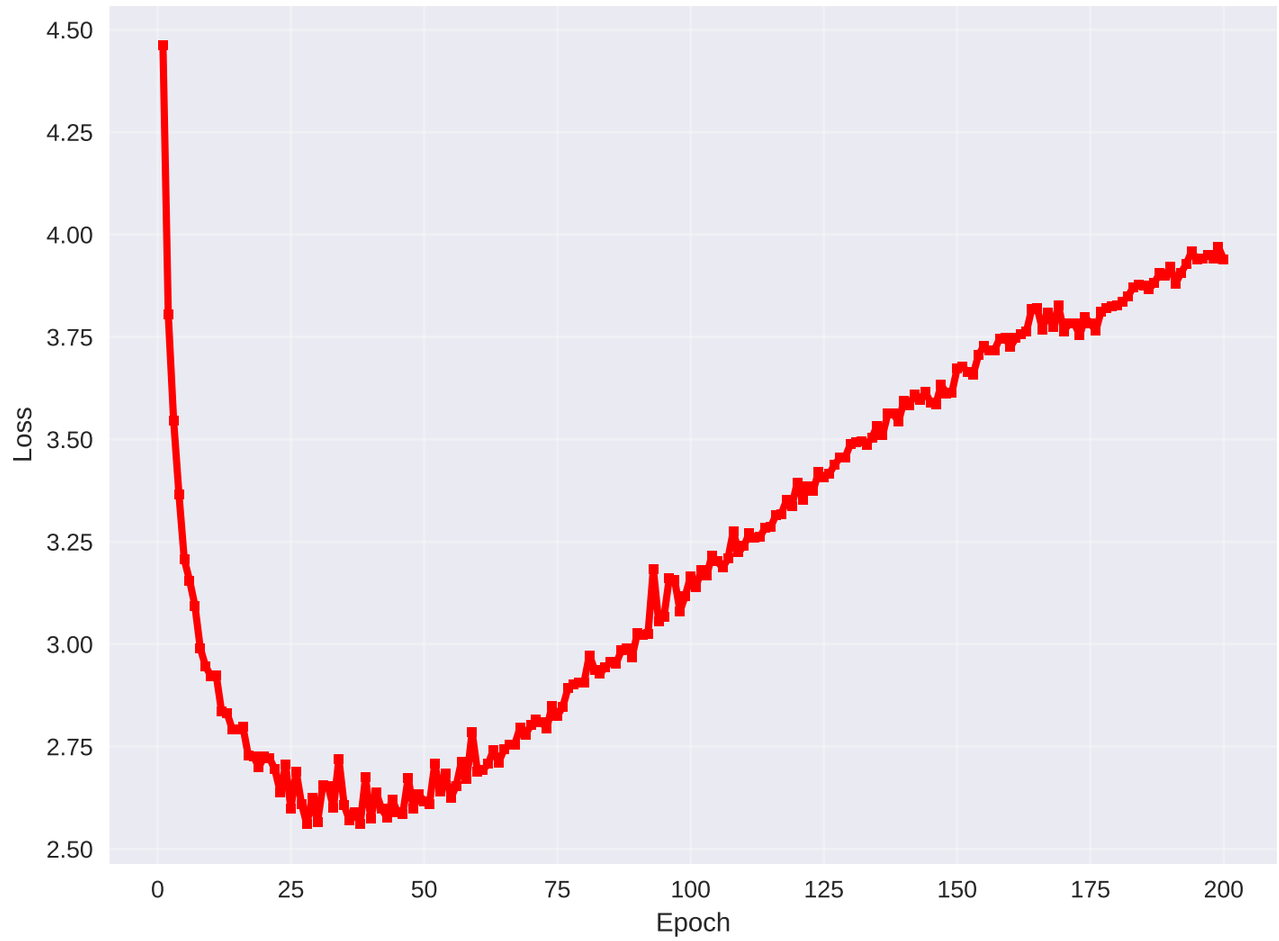
Federated Learning - Main Metrics Overview

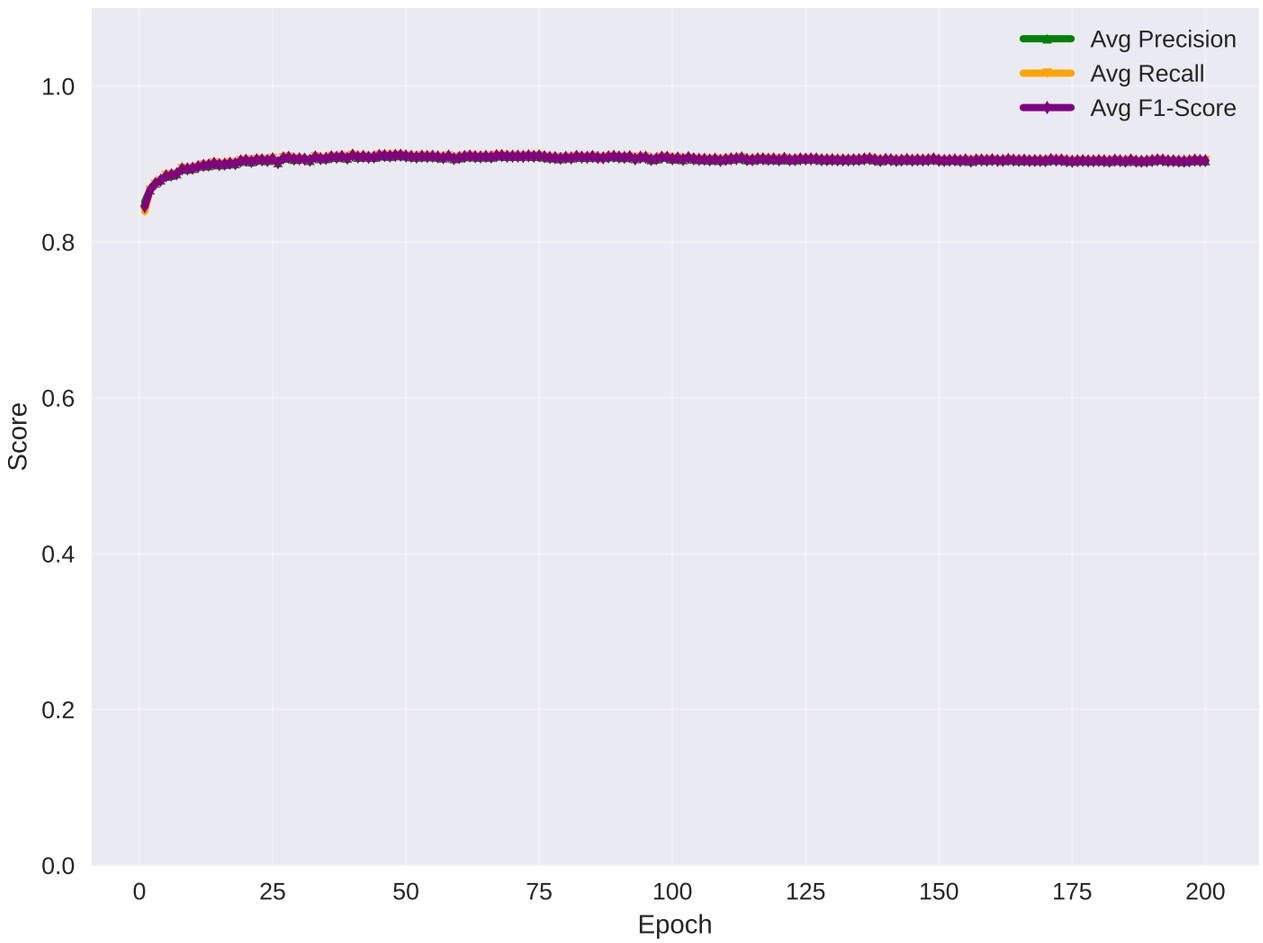
□ Global Model Accuracy Evolution



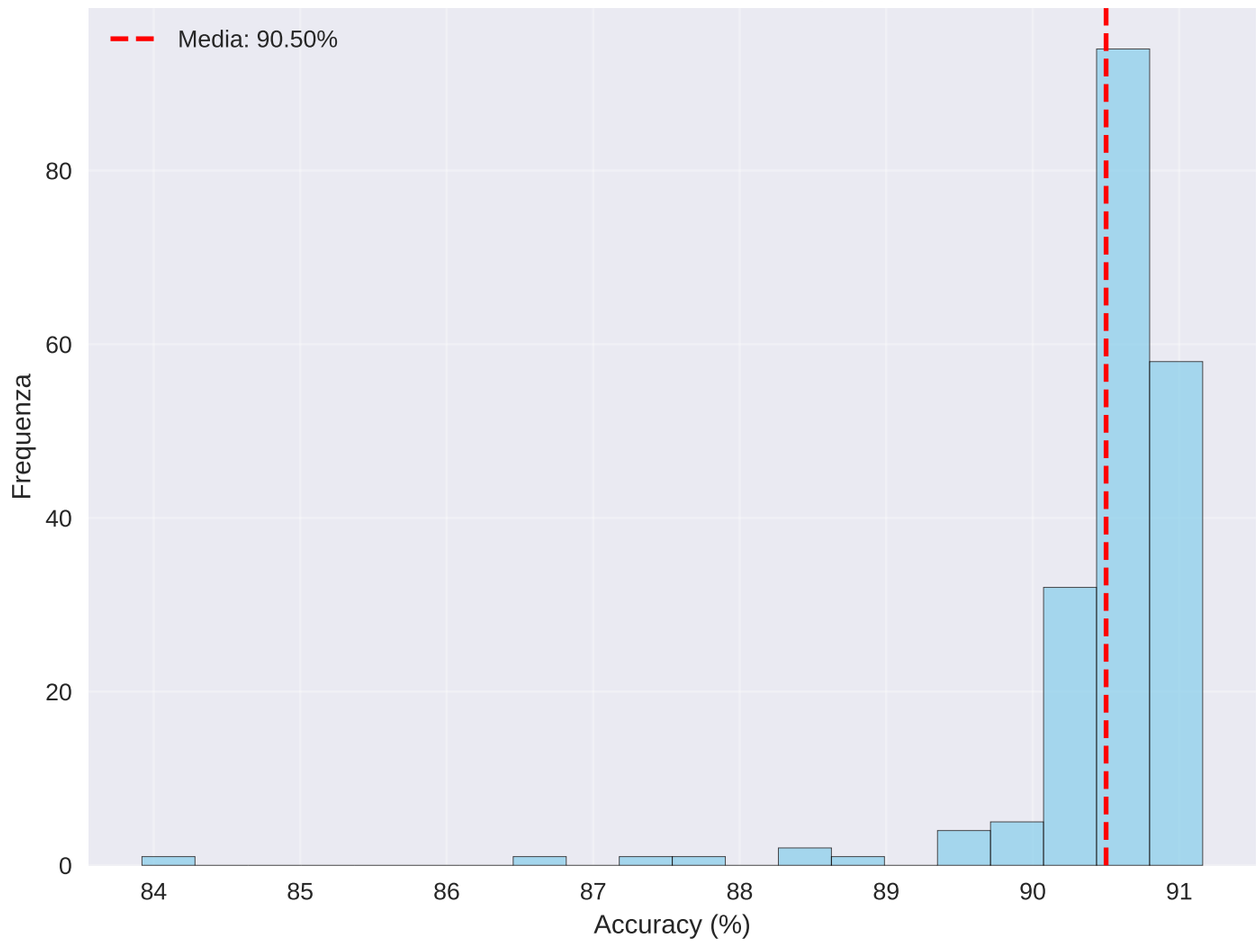
□ Global Model Loss Evolution



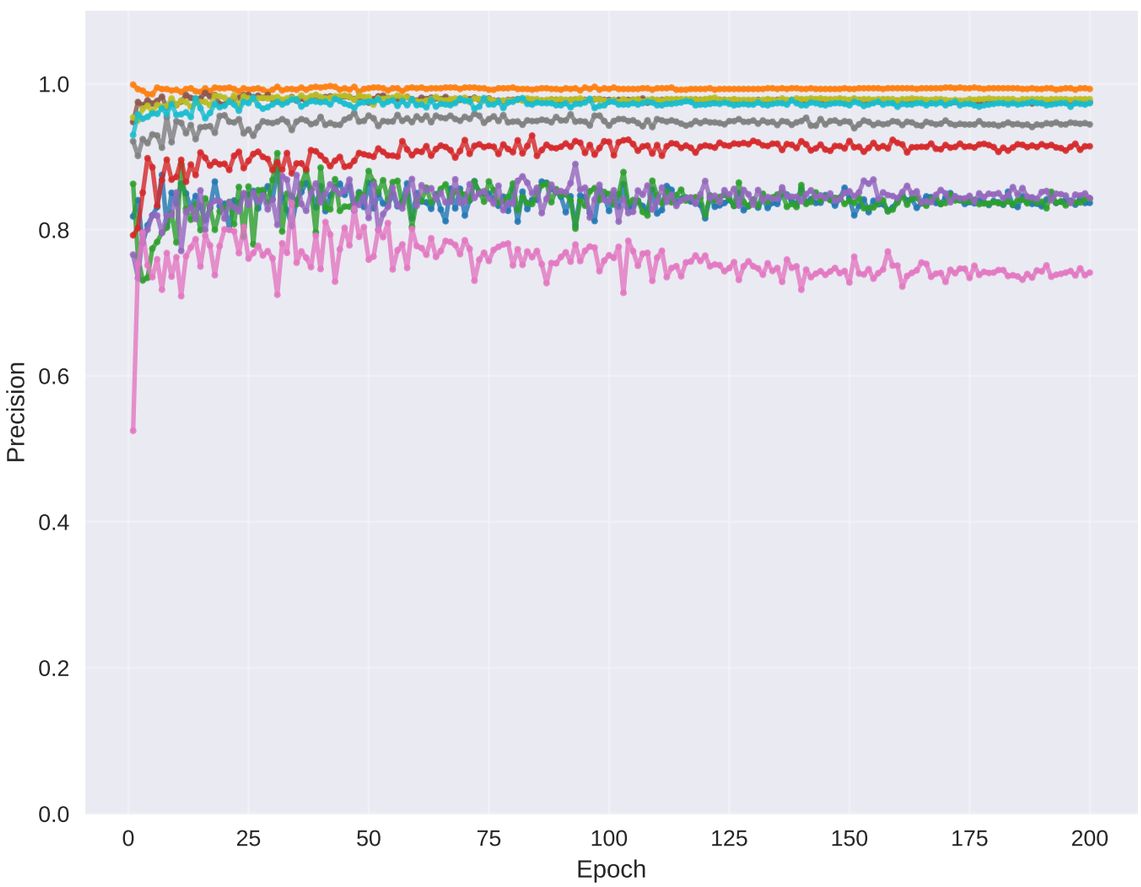
□ Average Precision, Recall & F1-Score



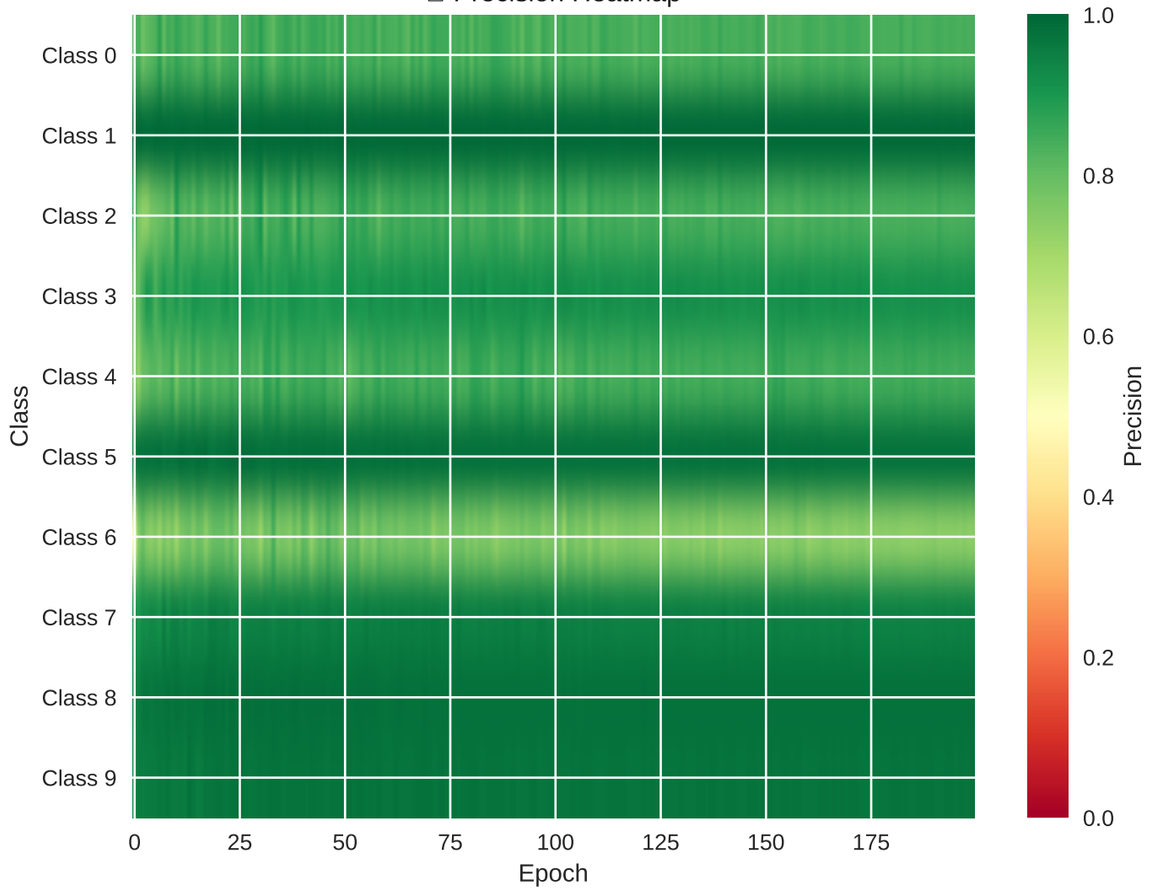
□ Distribuzione Accuracy

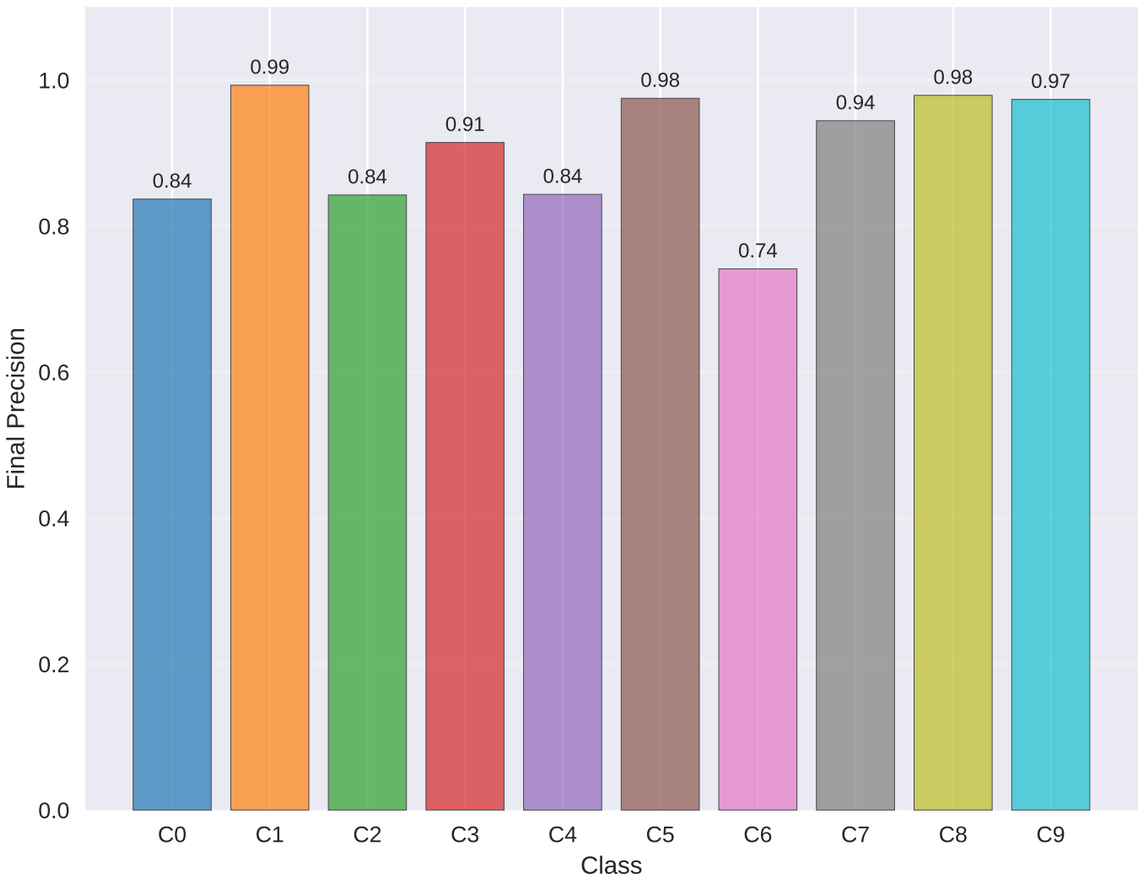


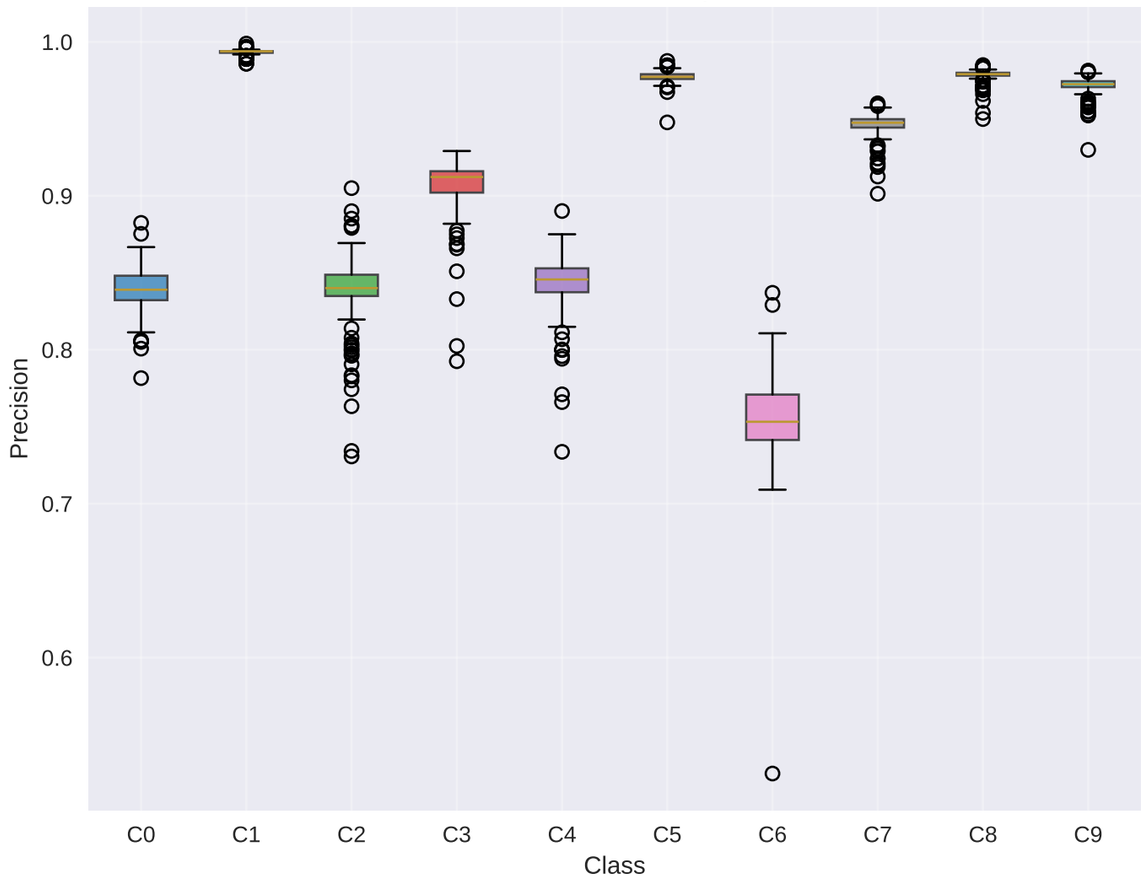
Per-Class Precision Analysis



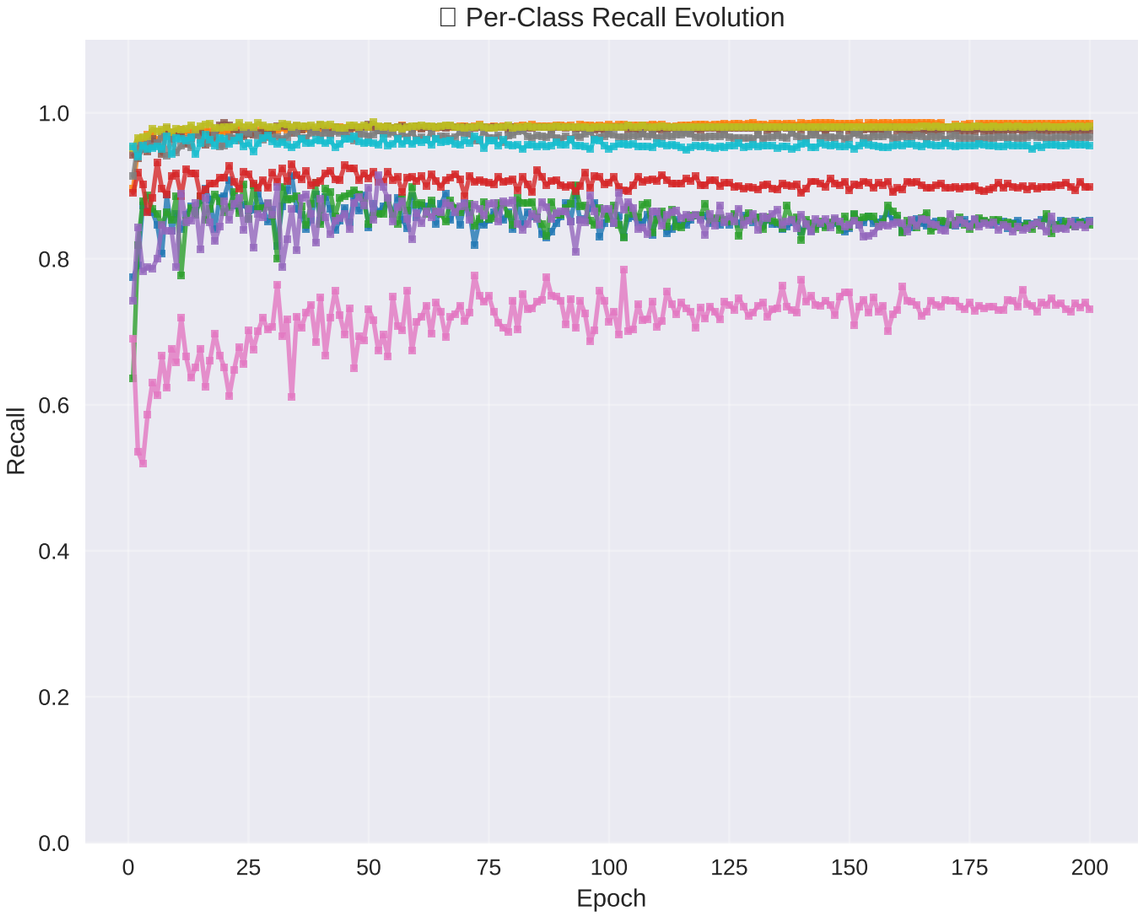


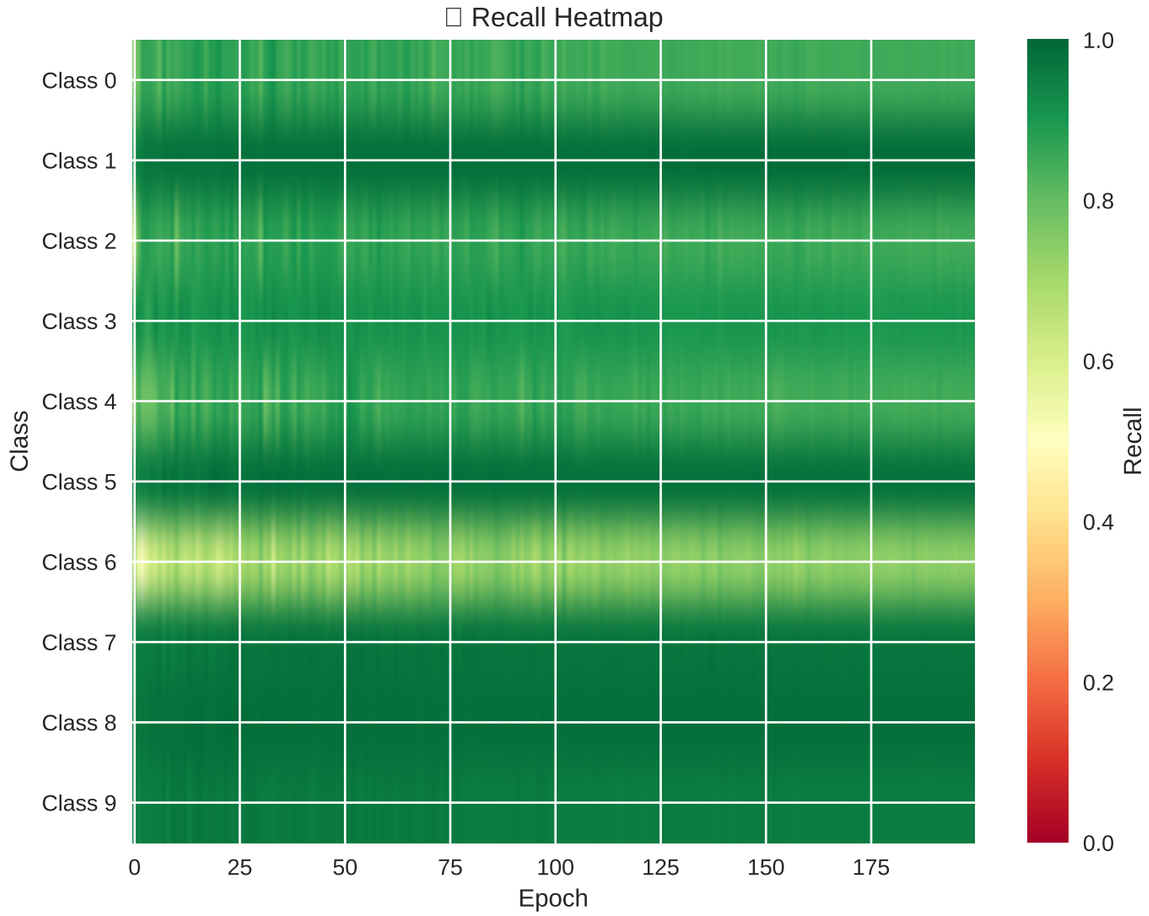


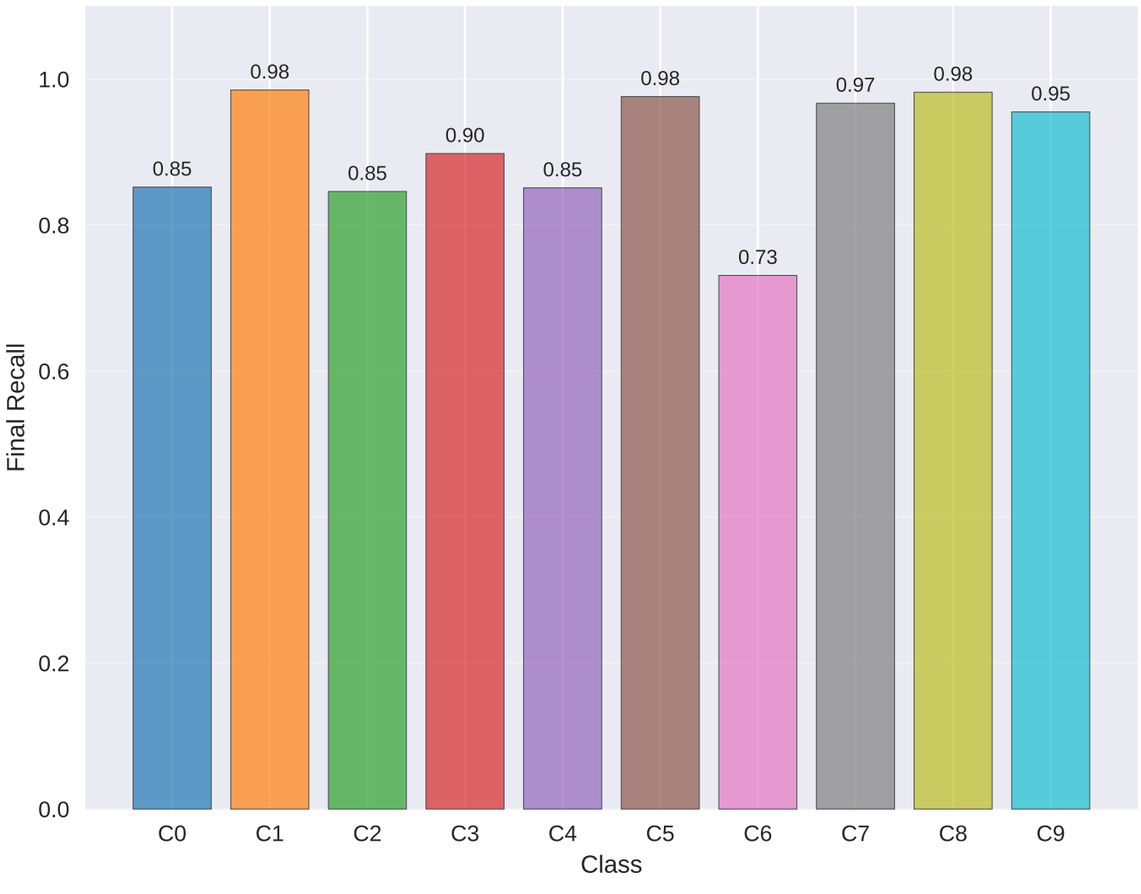


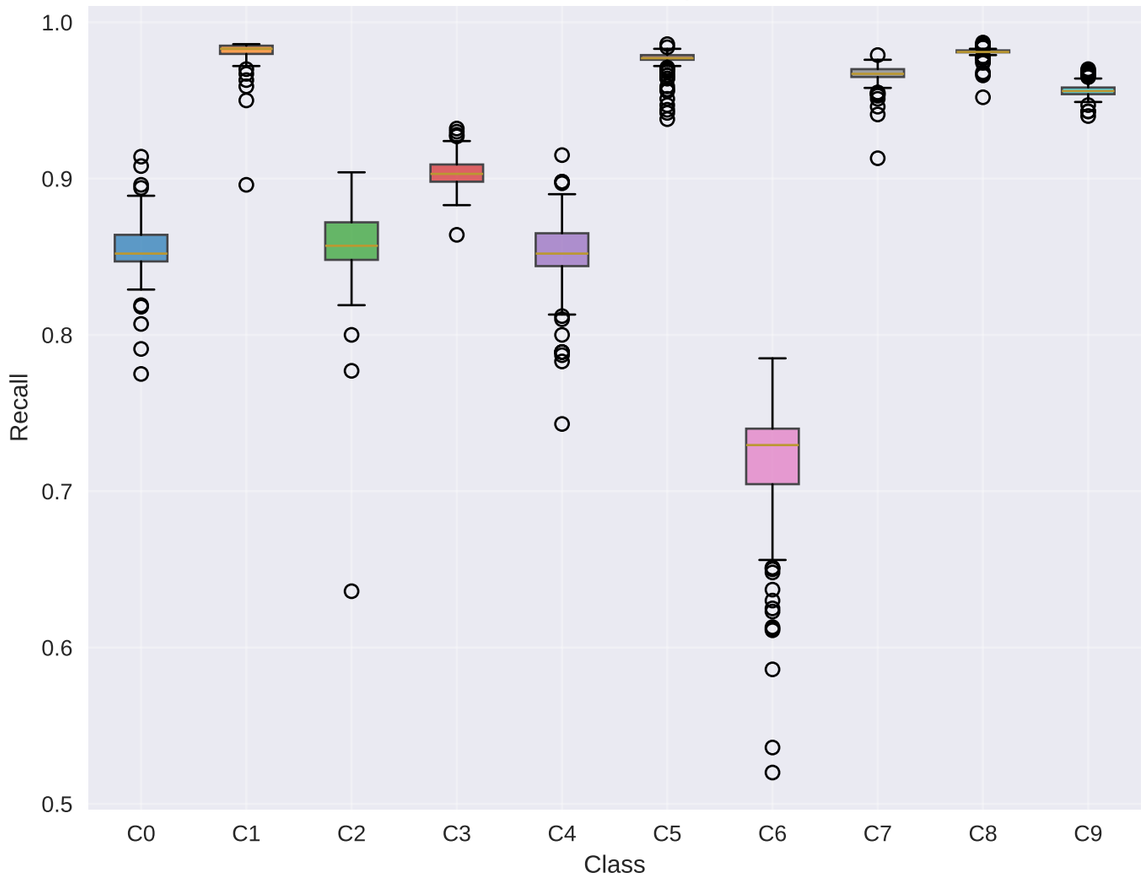


Per-Class Recall Analysis



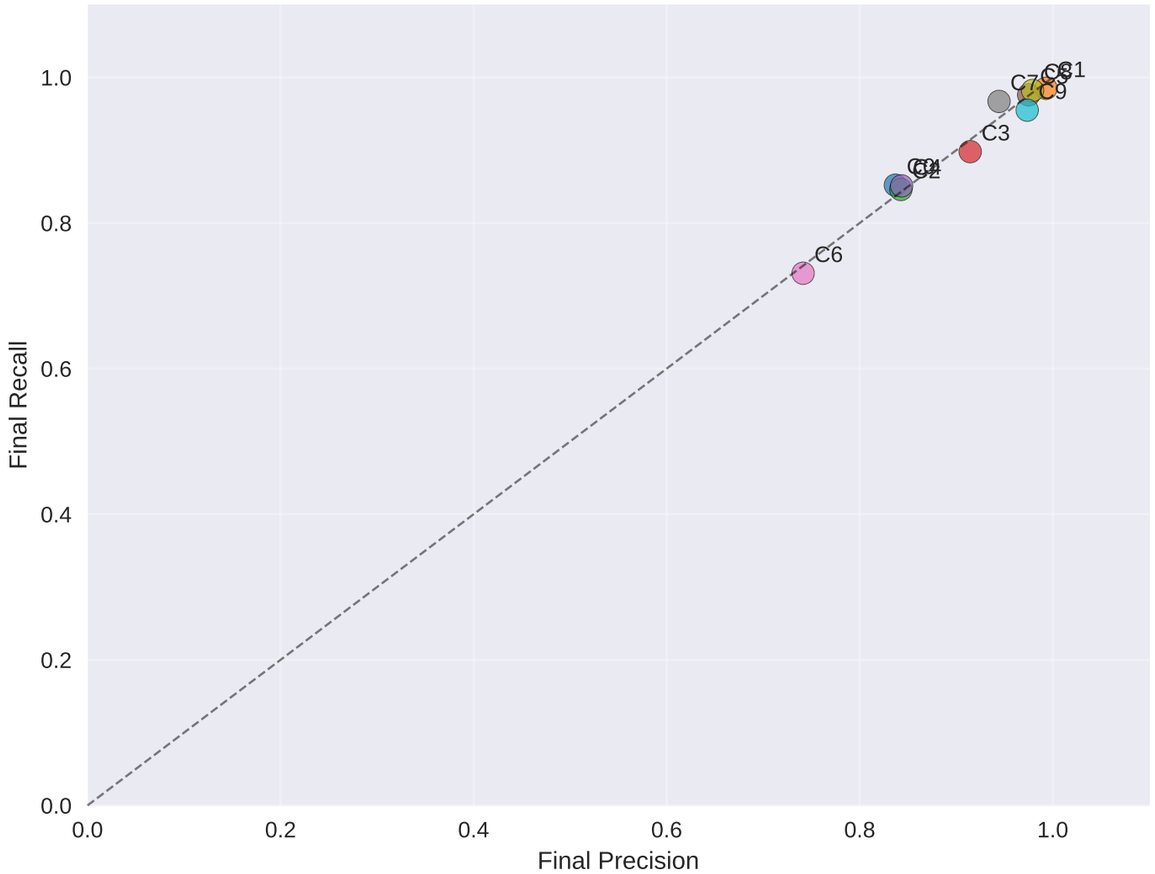




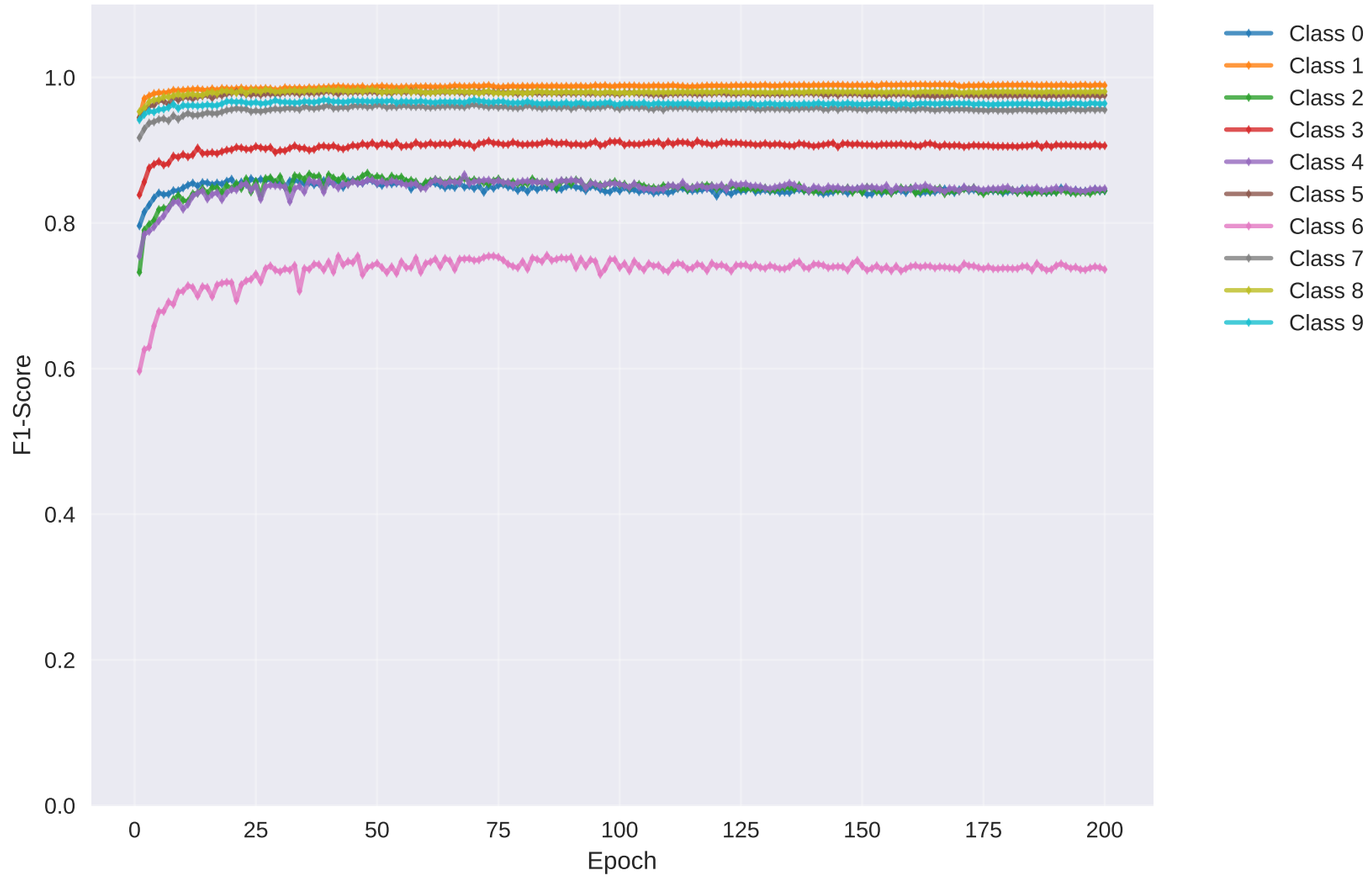


Comparative Analysis

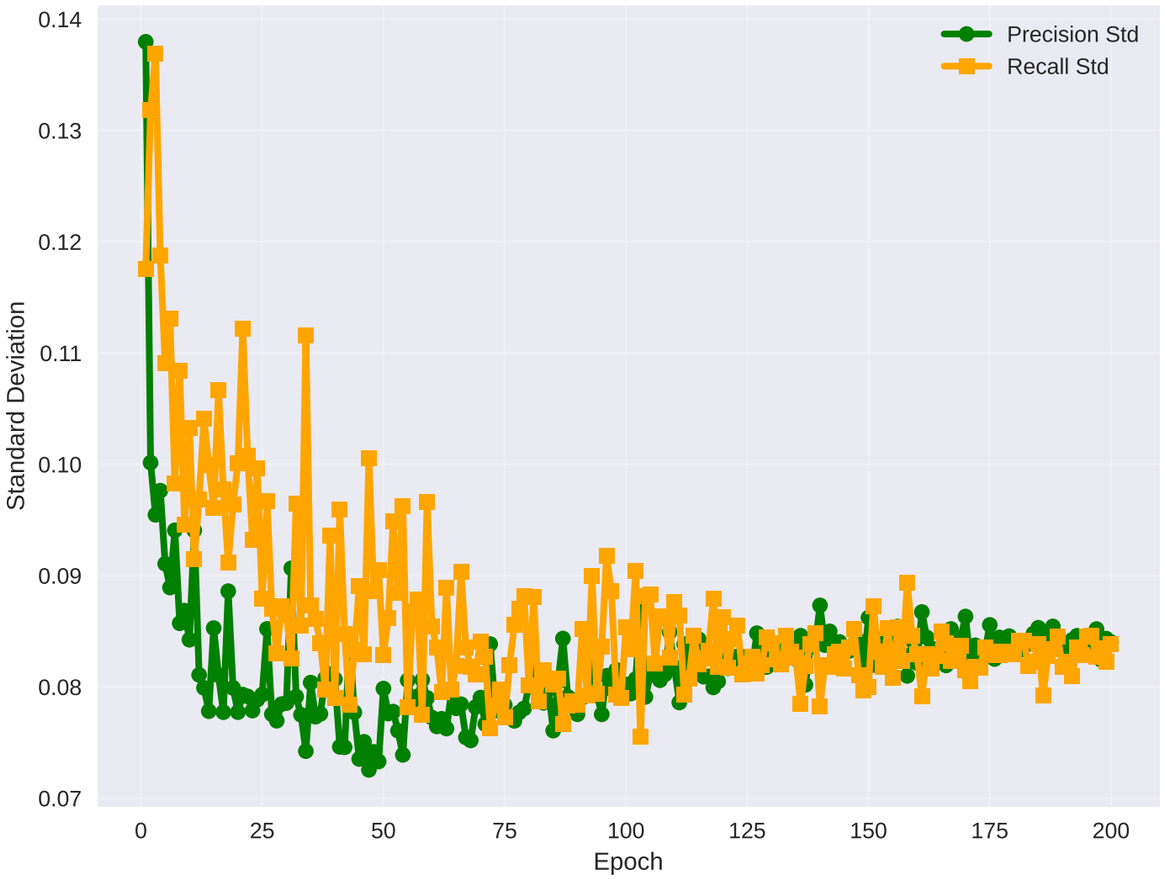
□ Precision vs Recall (Final Values)



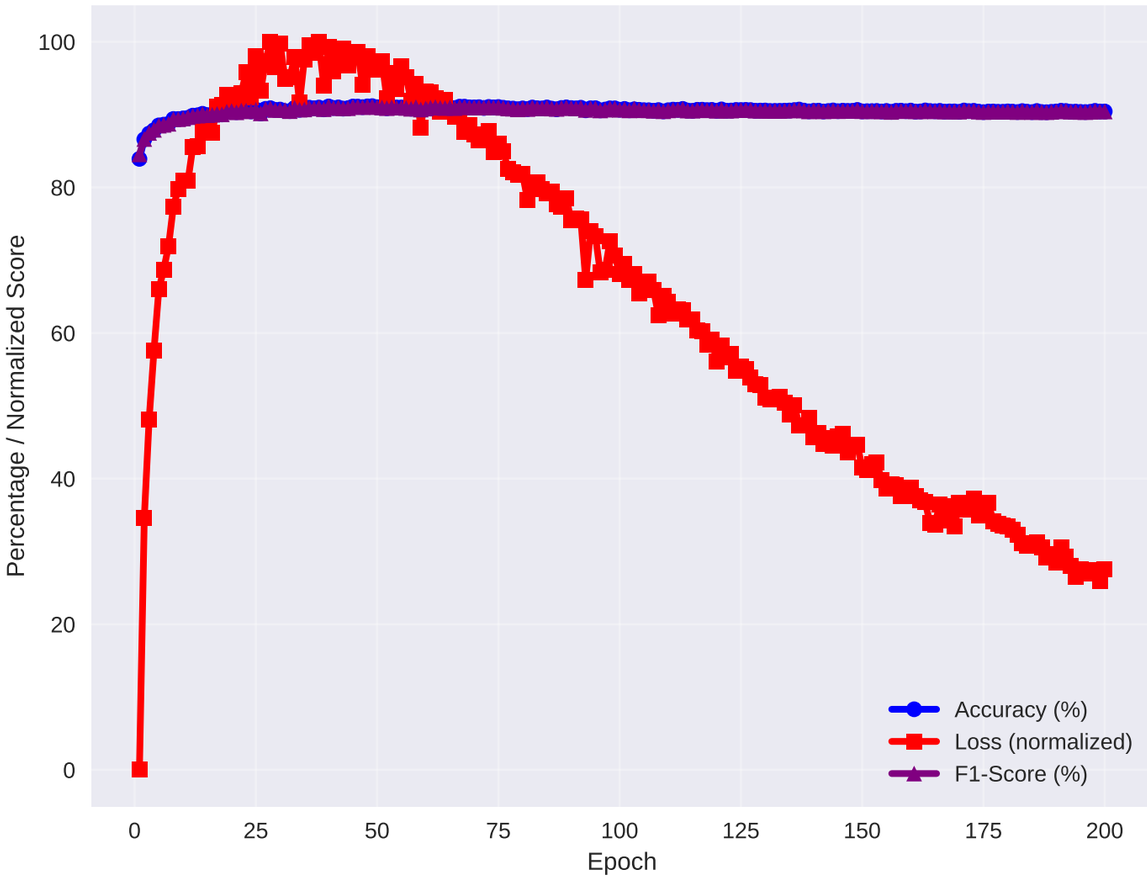
□ Per-Class F1-Score Evolution



□ Inter-Class Variance Evolution



□ Learning Curves Comparison



Performance Summary Report

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FEDERATED LEARNING - PERFORMANCE SUMMARY
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FILE ANALIZZATO: 3000_results.csv
DATA ANALISI: 2025-07-28 14:14:02
EPOCHE TOTALI: 200

METRICHE FINALI:
• Accuracy: 90.43%
• Loss: 3.9393
• Average Precision: 0.904
• Average Recall: 0.904
• Average F1-Score: 0.904

EVOLUZIONE TRAINING:
• Miglioramento Accuracy: +6.51%
• Riduzione Loss: +0.5232
• Stabilità Accuracy (CV): 0.0081

PERFORMANCE PER CLASSE:
• Migliore Precision: Classe 1 (0.993)
• Peggiore Precision: Classe 6 (0.741)
• Migliore Recall: Classe 1 (0.985)
• Peggiore Recall: Classe 6 (0.731)

ANALISI DETTAGLIATA PER CLASSE:
• Classe 0: P=0.837, R=0.852, F1=0.844
• Classe 1: P=0.993, R=0.985, F1=0.989
• Classe 2: P=0.843, R=0.846, F1=0.844
• Classe 3: P=0.914, R=0.898, F1=0.906
• Classe 4: P=0.843, R=0.851, F1=0.847
• Classe 5: P=0.975, R=0.976, F1=0.976
• Classe 6: P=0.741, R=0.731, F1=0.736
• Classe 7: P=0.944, R=0.967, F1=0.956
• Classe 8: P=0.979, R=0.982, F1=0.981
• Classe 9: P=0.973, R=0.955, F1=0.964

VALUTAZIONE QUALITATIVA:
• Performance Globale: BUONA
• Stabilità Training: STABILE
• Balance Classes: SBILANCIATO

RACCOMANDAZIONI:
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